Build and Run

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Learning Journey and Evidence

Prior Knowledge and Experience

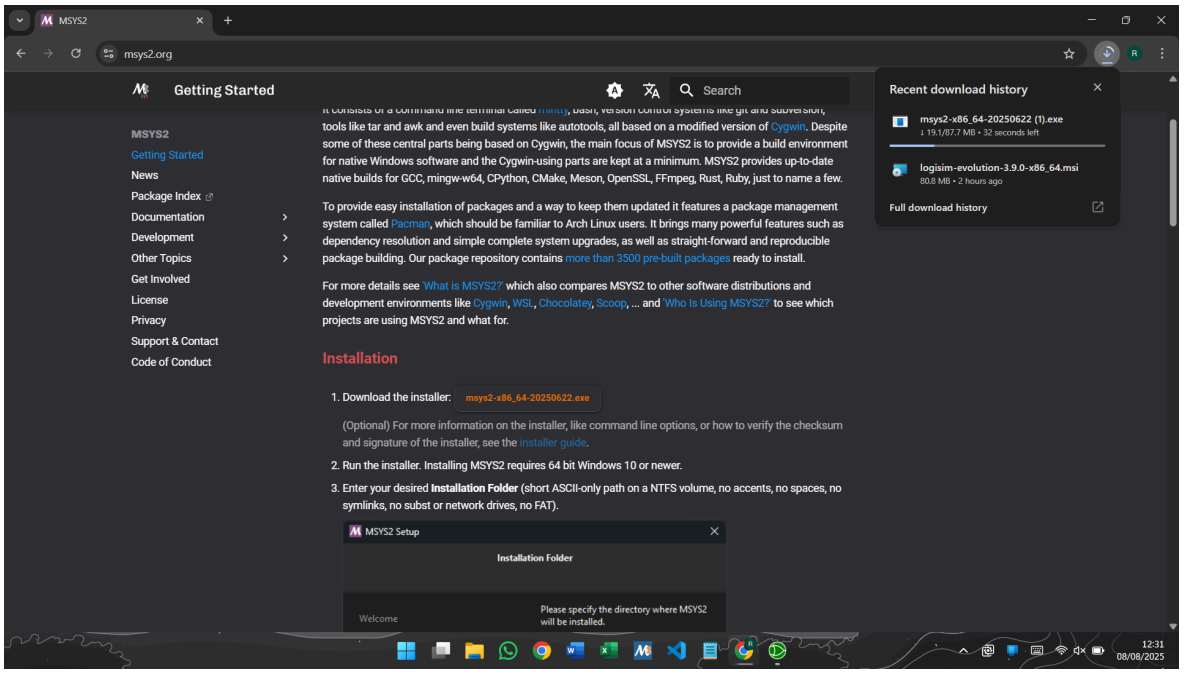
My journey started when i was 15 years old, i started coding in microsoft visual basic(console mode), writing simple programs then when i was nearly 17 years started coding python in VS code. My whole programming thus far has been fun and challenging and I am looking forward to continuing it.

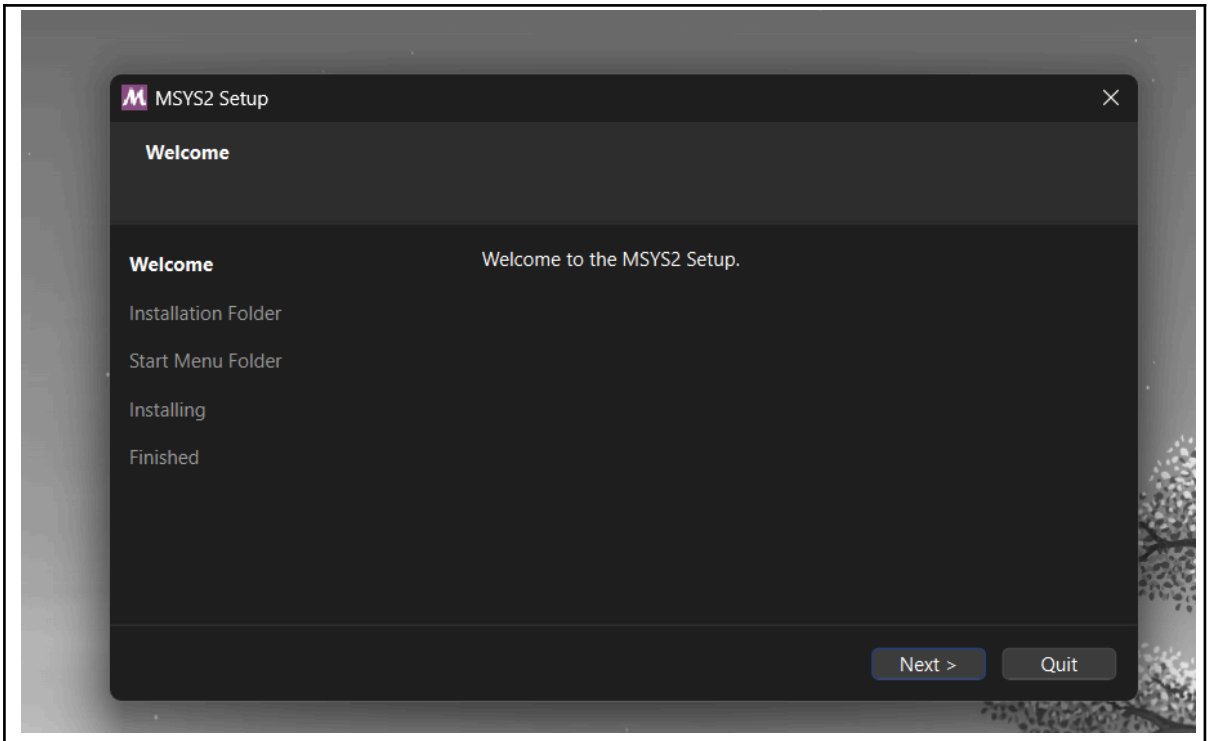
Study to Acquire, Refresh, and/or Extend Capability

To study and master this topic I did the following:

25/07/2025 - Read the Field guide
Read the field guide multiple times.
Understand what to do and what is expected of me.

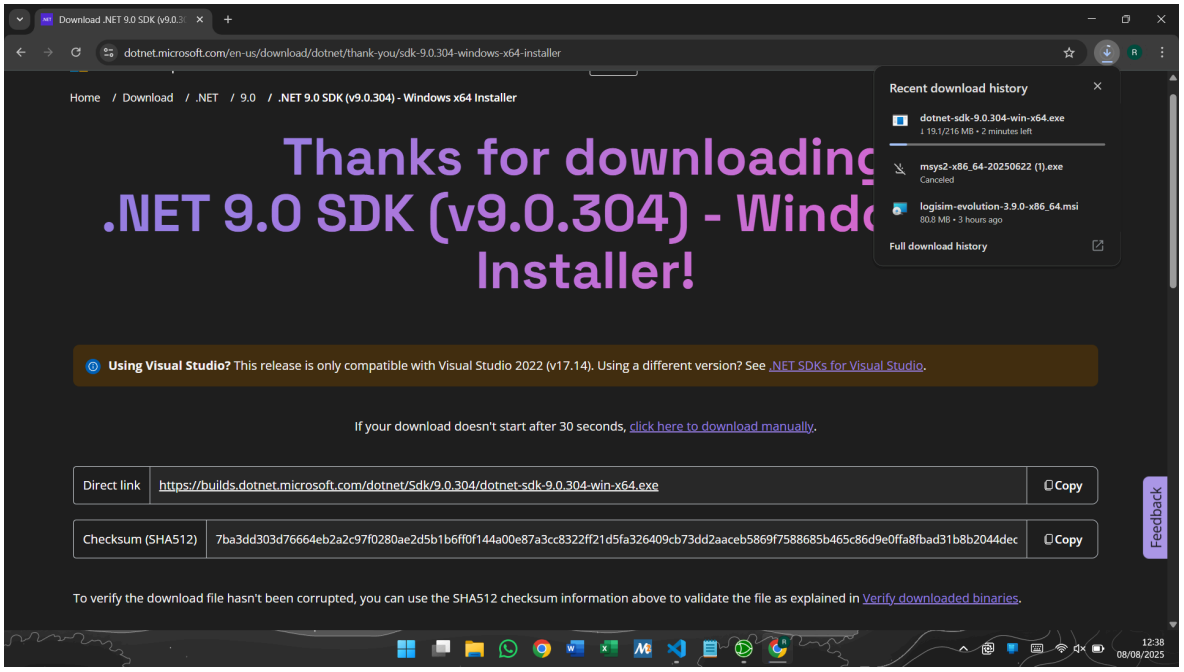
26/07/2025 - Download and install the required things
Vs code was already installed on my laptop; Thus I installed MSYS2, .NET and splashkit package.



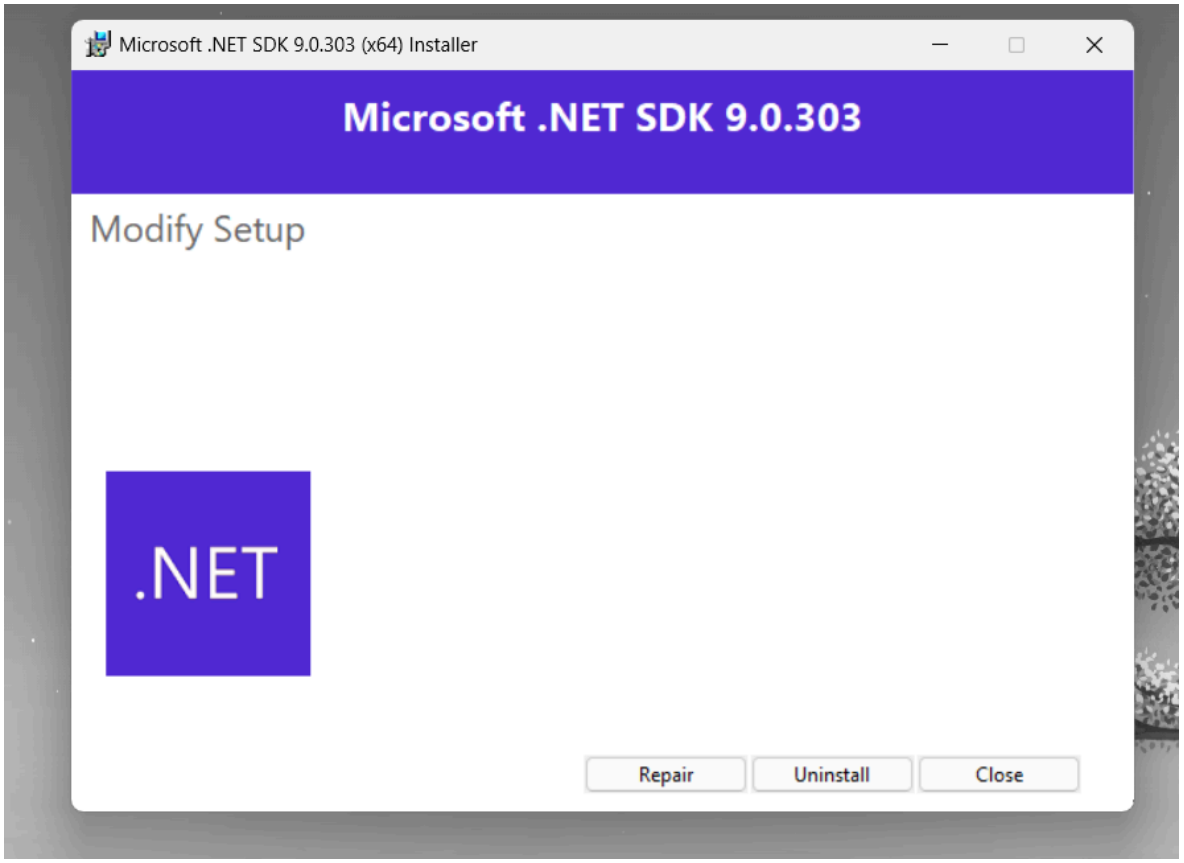


MSYS2 installed correctly:

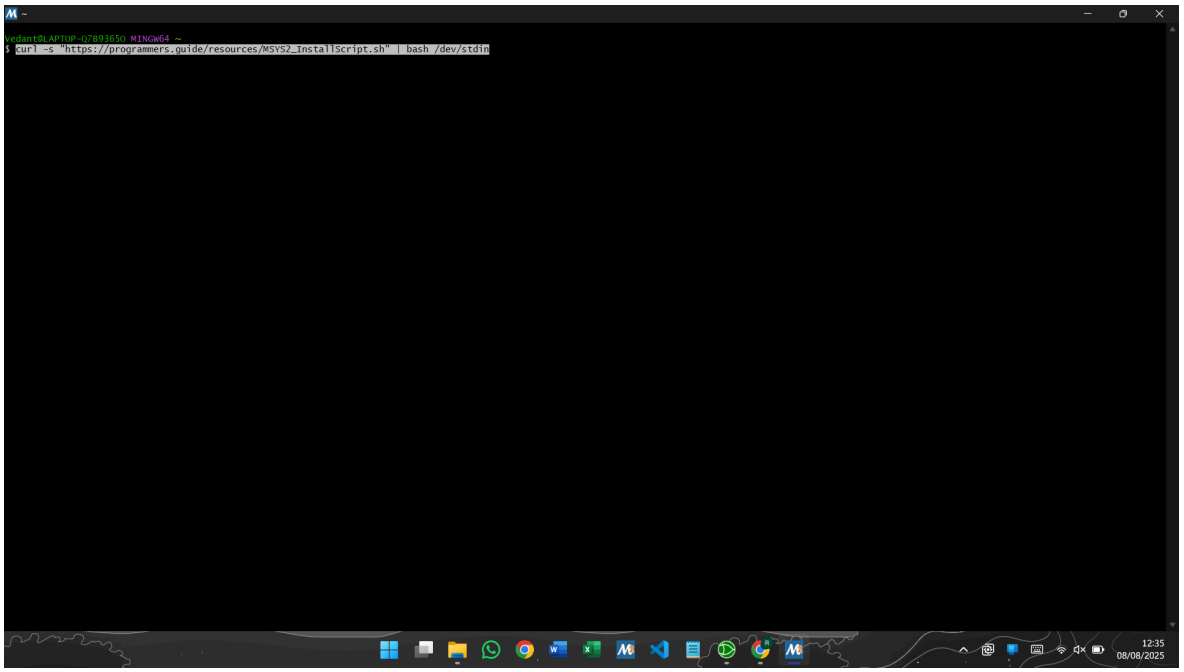
Downloaded .net:



.NET installed correctly:



Run the automated scrips:

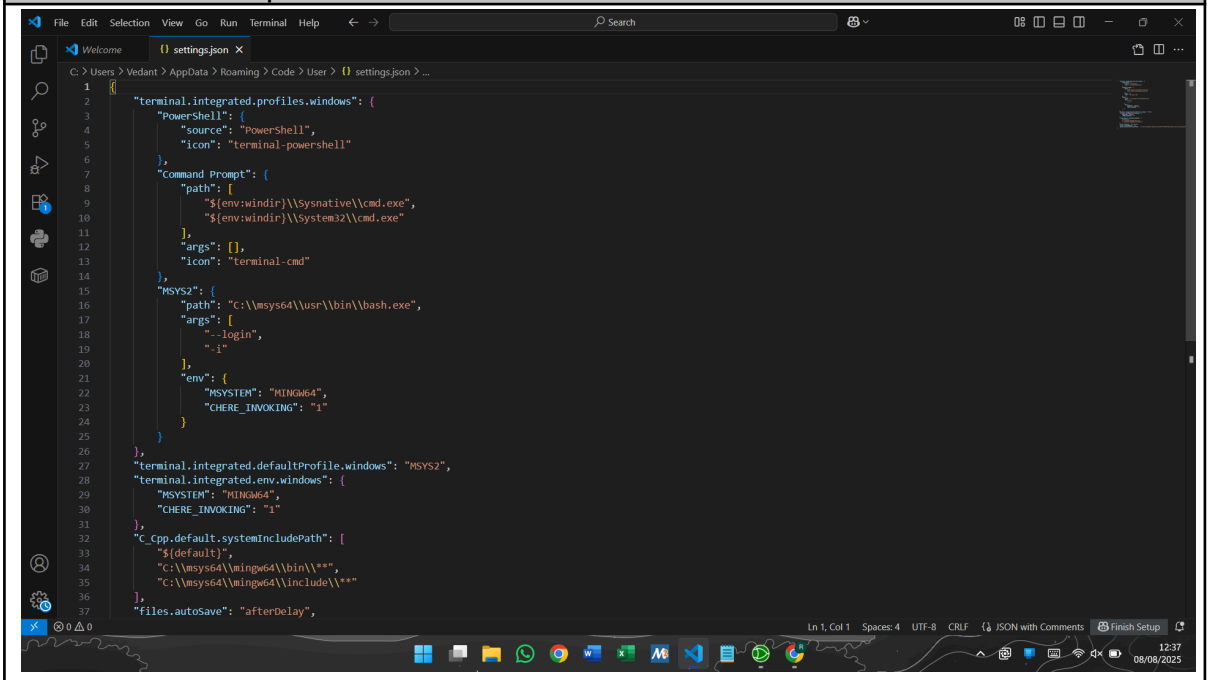


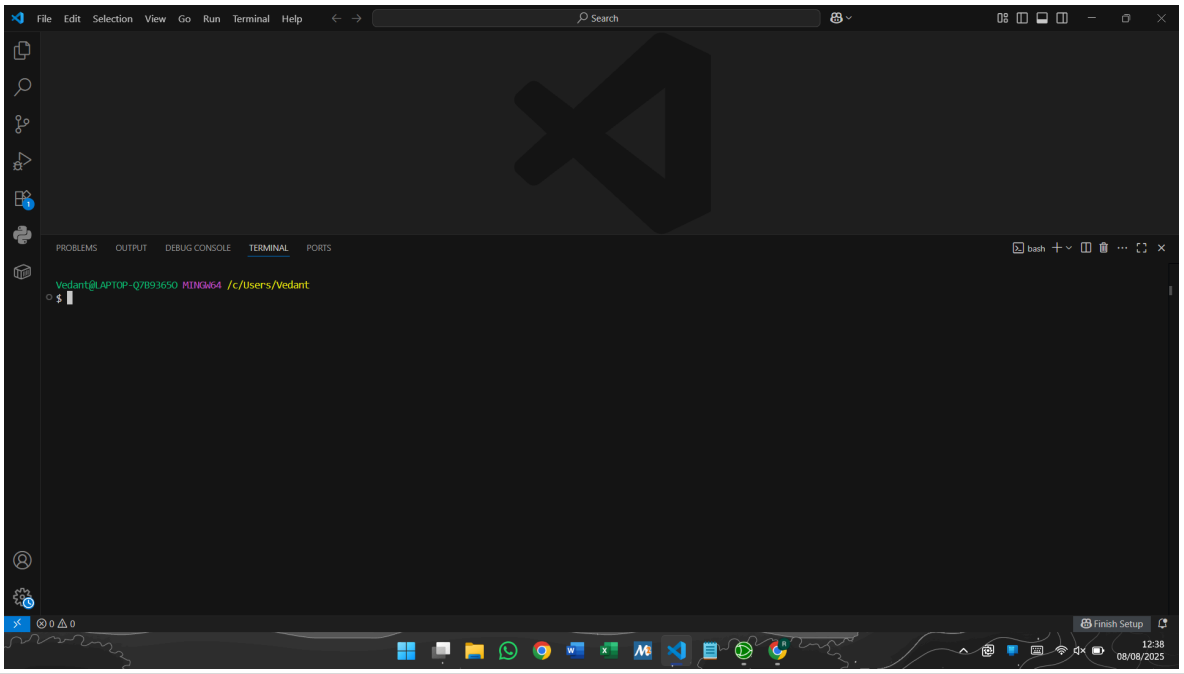
splashkit package installed successfully:

```
~

Vedant@LAPTOP-Q7B93650 MINGW64 ~
$ skm
Splashkit is installed successfully!
Missing skm command. For help use 'skm help'

Vedant@LAPTOP-Q7B93650 MINGW64 ~
$ |
```

Successfully installed.

25/07/2025 - setup MSYS2 terminal in vs code





Successfully setup VScode terminal.

26/07/2025 - MSYS2 MINGW64 shell

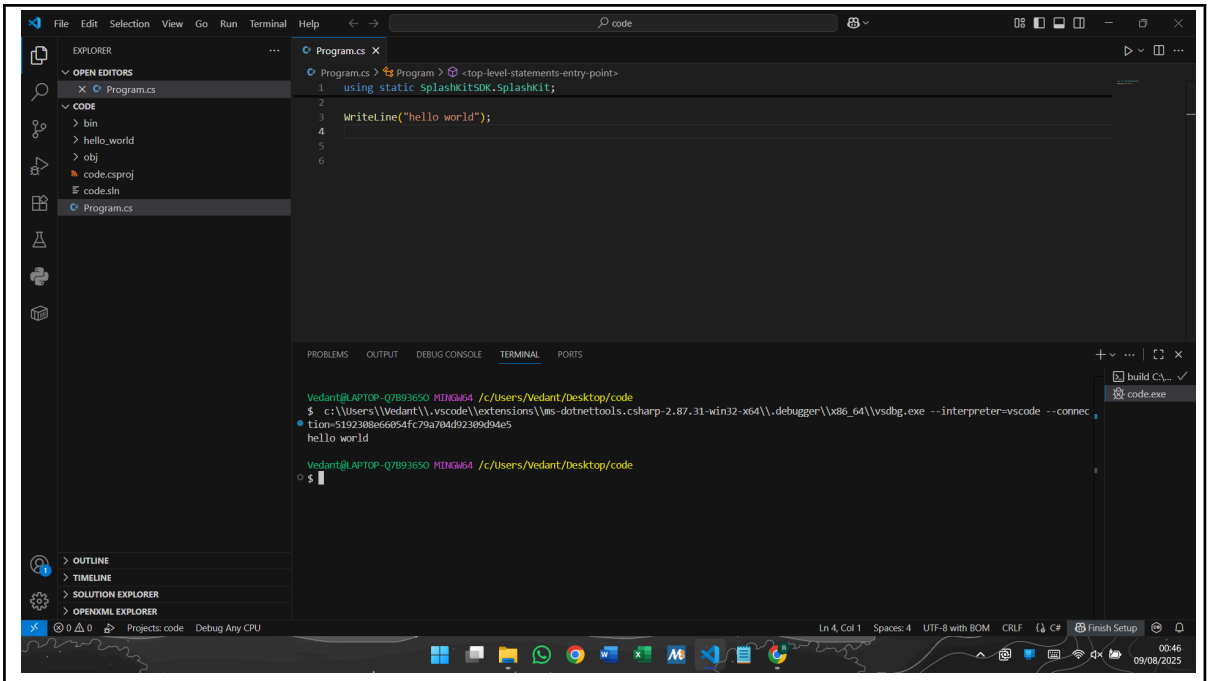
```
/c/Users/Vedant/Desktop/mycode/newfolder

Vedant@LAPTOP-Q7B93650 MINGW64 ~
$ cd /c/Users/Vedant/Desktop/mycode

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/mycode
$ mkdir newfolder

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/mycode
$ ls
HelloWorld  MyCppCode  newfolder

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/mycode
$ cd newfolder

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/mycode/newfolder
$ ls

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/mycode/newfolder
$ mkdir new_newfolder

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/mycode/newfolder
$ ls
new_newfolder

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/mycode/newfolder
$
```

i understood how to use shell to create, list, delete folders.

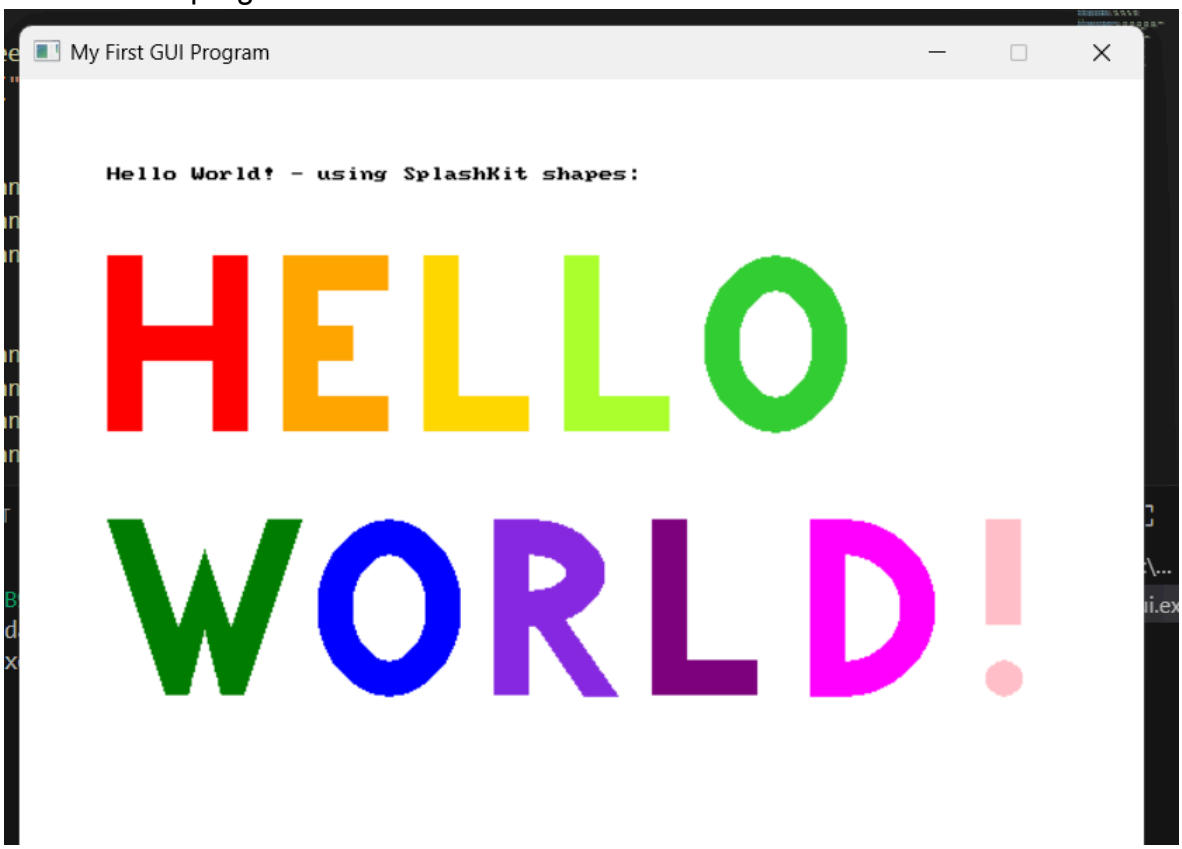
26/07/2025 - C# file

created a folder “code” to store all my code:

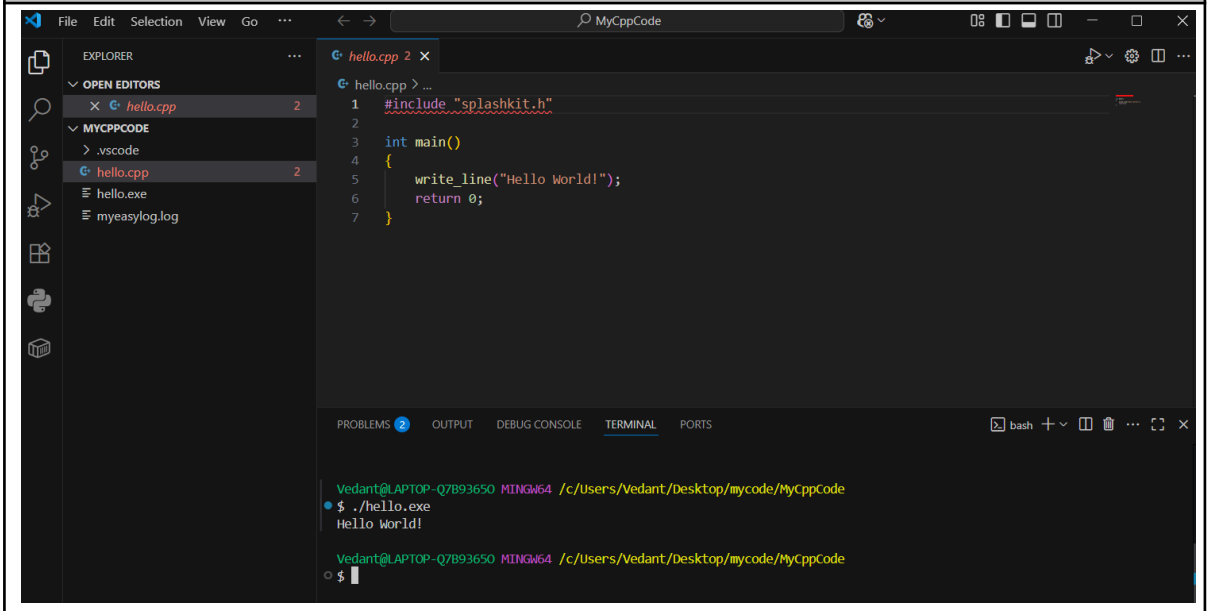
```
/c/Users/Vedant/Desktop/code
Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop
$ mkdir code

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop
$ cd code

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/code
$ mkdir hello_world

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/code
$ dotnet new console
The template "Console App" was created successfully.

Processing post-creation actions...
Restoring C:\Users\Vedant\Desktop\code\code.csproj:
Restore succeeded.


Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/code
$ dotnet add package splashkit

Build succeeded in 0.7s
info : X.509 certificate chain validation will use the default trust store selected by .NET for code
 signing.
info : X.509 certificate chain validation will use the default trust store selected by .NET for time
stamping.
info : Adding PackageReference for package 'splashkit' into project 'C:\Users\Vedant\Desktop\code\co
de.csproj'.
info :   GET https://api.nuget.org/v3/registration5-gz-semver2/splashkit/index.json
info :   OK https://api.nuget.org/v3/registration5-gz-semver2/splashkit/index.json 238ms
info : Restoring packages for C:\Users\Vedant\Desktop\code\code.csproj...
info :   GET https://api.nuget.org/v3/vulnerabilities/index.json
info :   OK https://api.nuget.org/v3/vulnerabilities/index.json 285ms
info :   GET https://api.nuget.org/v3-vulnerabilities/2025.07.31.17.40.39/vulnerability.base.json
info :   GET https://api.nuget.org/v3-vulnerabilities/2025.07.31.17.40.39/2025.08.08.11.41.12/vulner
ability.update.json
info :   OK https://api.nuget.org/v3-vulnerabilities/2025.07.31.17.40.39/vulnerability.base.json 222
ms
info :   OK https://api.nuget.org/v3-vulnerabilities/2025.07.31.17.40.39/2025.08.08.11.41.12/vulnera
bility.update.json 260ms
info : Package 'splashkit' is compatible with all the specified frameworks in project 'C:\Users\Veda
nt\Desktop\code\code.csproj'.
info : PackageReference for package 'SplashKit' version '1.2.12' added to file 'C:\Users\Vedant\Desk
top\code\code.csproj'.
info : Writing assets file to disk. Path: C:\Users\Vedant\Desktop\code\obj\project.assets.json
log  : Restored C:\Users\Vedant\Desktop\code\code.csproj (in 723 ms).

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/code
$ code .

Vedant@LAPTOP-Q7B93650 MINGW64 /c/Users/Vedant/Desktop/code
$
```



The GUI test program runs as well:



I understood how to create, code and run C# file

26/07/2025 - c/c++

compiled using the the clang++ hello.cpp -o hello -l splashkit
I understood how to create, code and run c/c++ file

Brief Summary of Concepts

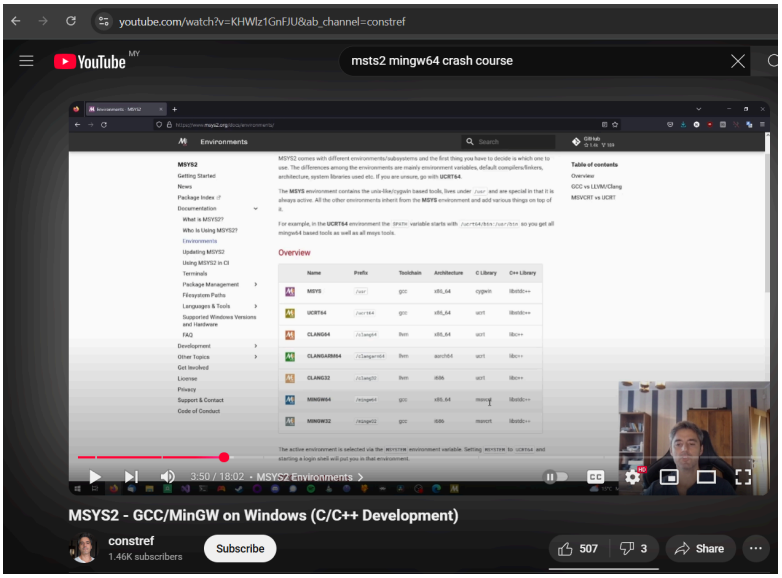
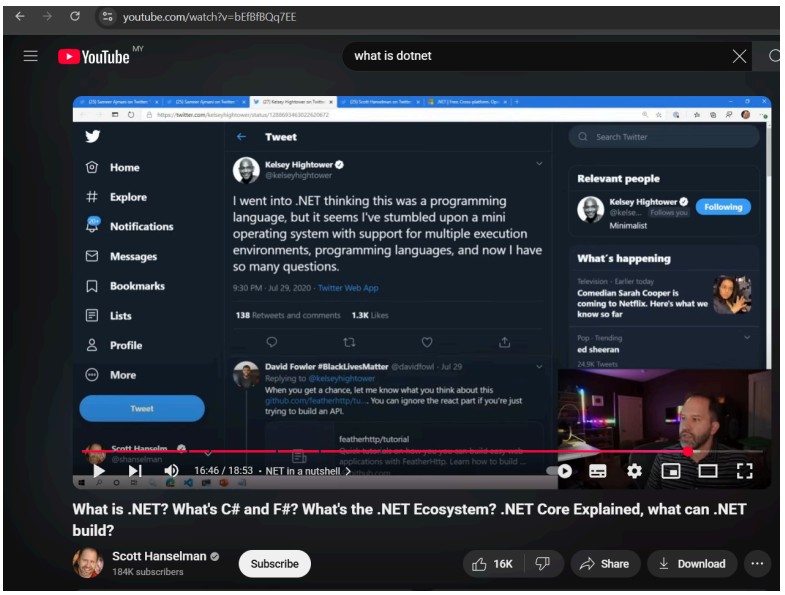
Concept / Commands	Key Idea / Concept
Compiler	translate the whole program in machine code so the machine can run it as well as creating an error list.
Shell	It is a program that allows us to run commands
Terminal	An interface that allows us to interact with the computer

Action	Example shell command
Make a folder	`mkdir`
Move into a folder	`cd`
Show/list the files in a folder	`ls`
Create a dotnet project	`dotnet new console`
Add SplashKit to a dotnet project	`add package splashkit`
Build and run a dotnet project	`dotnet run`
Compile a C/C++ program – which uses the SplashKit library	`clang++ filename.ccp -o appname -l splashkit`
Run the C++ program	`./appname.exe`

Reflection

What is the most important thing you learned from this and why?

What I consider to be the most important thing that I learned is the syntax of C/C++ and C# projects and how to compile and run them. As I have never code in these languages before, I needed this knowledge to be able to make use of my existing programming/problem solving experience from other programming languages to start building programs.

Using the terminal can be quite challenging. What strategies are you going to put in place to help you learn to use this?

I will do research, as well as ‘mess around’ with it to familiarize myself and gain experience on how to properly use the terminal.
watch some youtube videos: